

CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · PYQ SOLUTIONS

Chapter 2: Banking (Recurring Deposit Accounts)

Step-by-step solutions to the previous year questions

The question numbers match the PYQ question sheet.

1 Interest and maturity value

1. A man deposits ₹500 every month for 18 months in a recurring deposit account and earns ₹570 as interest. Which expression gives the maturity amount? (a) ₹(500 × 18 + 570) (b) ₹(500 × 19 + 570) (c) ₹(500 × 18 × 19 + 570) (d) ₹(500 × 9 × 19 + 570) **ICSE 2025, 1 mark**

SOLUTION

Maturity amount = total money deposited + interest.

Total deposited = 500 × 18.

Answer: (a)

2 Finding the rate of interest

2. A woman deposits ₹250 a month in a recurring deposit account for 3 years and receives ₹10,110 on maturity. Find (a) the rate of interest (b) how much more interest she would earn by depositing ₹50 more each month, at the same rate and for the same time. **ICSE 2025, 4 marks**

SOLUTION

(a) $P = 250, n = 36$. Total deposited = $250 \times 36 = 9000$, so the interest is $10110 - 9000 = 1110$.

$$I = P \times \frac{n(n+1)}{2 \times 12} \times \frac{r}{100}; 1110 = 250 \times \frac{36 \times 37}{24} \times \frac{r}{100} = 138.75r$$

$$r = \frac{1110}{138.75} = 8. \text{ The rate is 8\% per annum.}$$

(b) The extra ₹50 a month earns $50 \times \frac{36 \times 37}{24} \times \frac{8}{100} = 50 \times 55.5 \times 0.08 = 222$.

Answer: (a) 8% per annum (b) ₹222 more

3 Finding the time

3. A man deposits ₹2000 a month in a recurring deposit account that pays 8% per annum. He receives ₹1040 as interest on maturity. For how many years was the account held?

ICSE 2024, 3 marks

SOLUTION

$$I = P \times \frac{n(n+1)}{2 \times 12} \times \frac{r}{100}; 1040 = 2000 \times \frac{n(n+1)}{24} \times \frac{8}{100}$$

$$1040 = \frac{20}{3} n(n+1), \text{ so } n(n+1) = 156 = 12 \times 13$$

$$n = 12 \text{ months}$$

Answer: 1 year

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